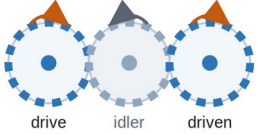
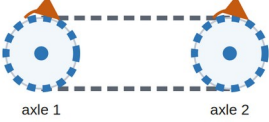
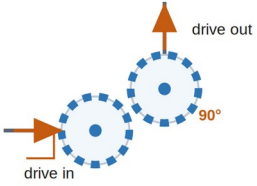
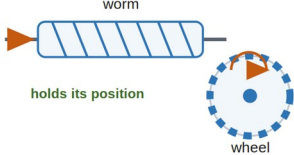
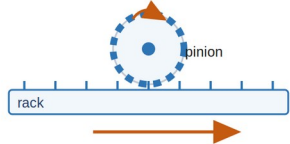
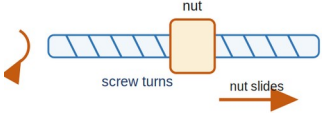
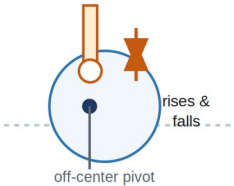
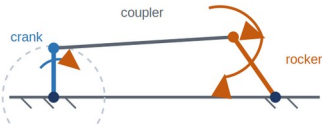


Advanced Mechanisms — Bench Card

Keep this at your build station

A quick reference for the advanced mechanisms in your kit — what each one does and when to reach for it. *For the full explanation of any mechanism (how it works, the heads-up, the build), see the Advanced Mechanisms reference on the course site.*

 Idler gear Bridges a gap; drive & driven turn the SAME way. Good for: 4-wheel drive	 Chain & sprocket Links two axles across a gap, same direction. Good for: 4WD / far-off shafts	 Bevel gear Turns the drive 90° onto a new axis. Good for: around-a-corner drives
 Differential One input → two wheels at different speeds. Good for: smooth powered turns	 Worm & wheel Big reduction + self-locking (holds its position). Good for: lifts & aimers that stay put	 Rack & pinion Rotation → straight-line motion. Good for: pushers, grippers, slides
 Leadscrew Spin the screw, the nut slides — slow, strong, holds. Good for: precise lifts	 Cam & follower Rotation → up-and-down (throw = 2 × offset). Good for: taps, beats, bobs	 Crank & rocker Rotation → back-and-forth swing. Good for: wipers, waving arms